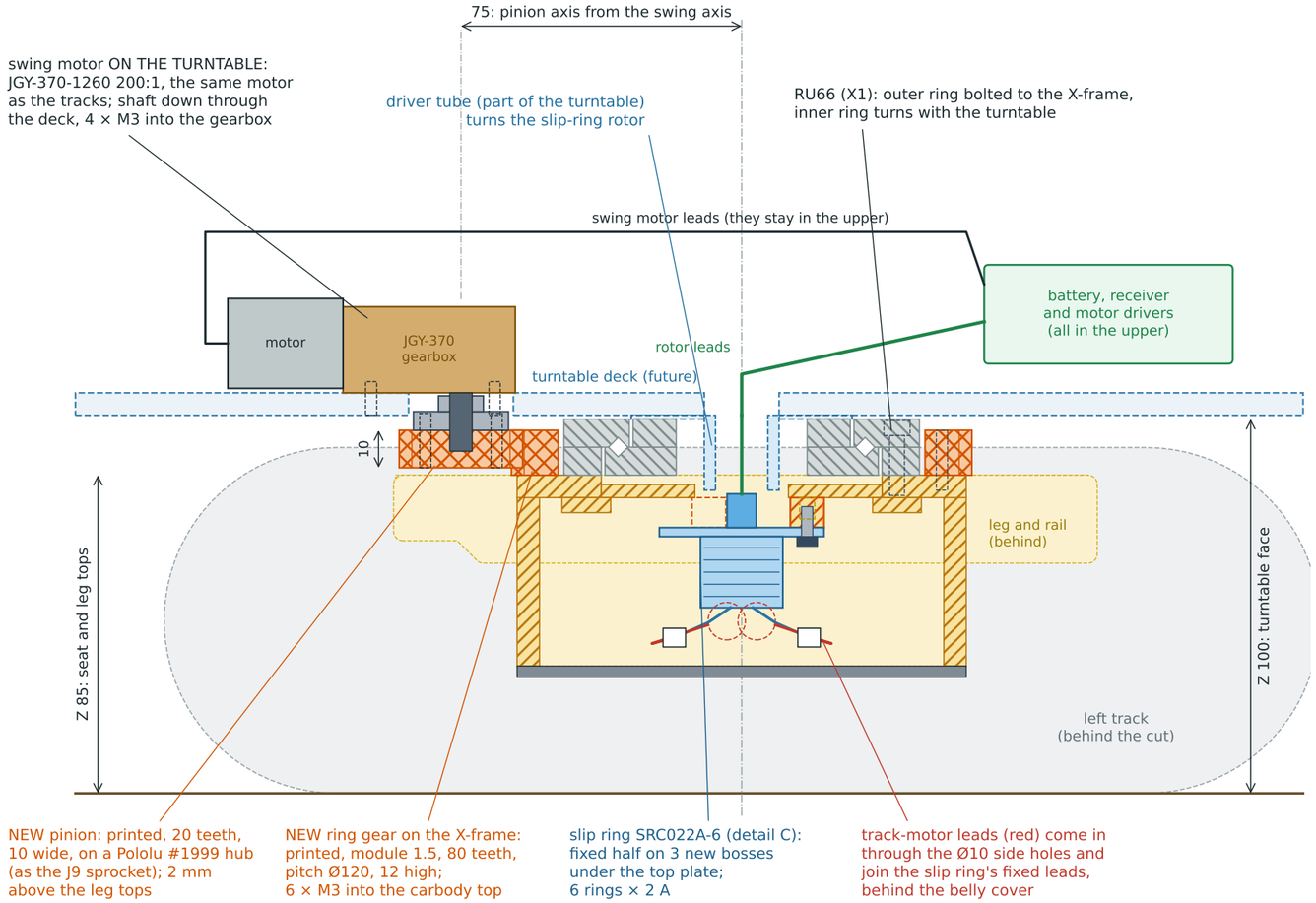
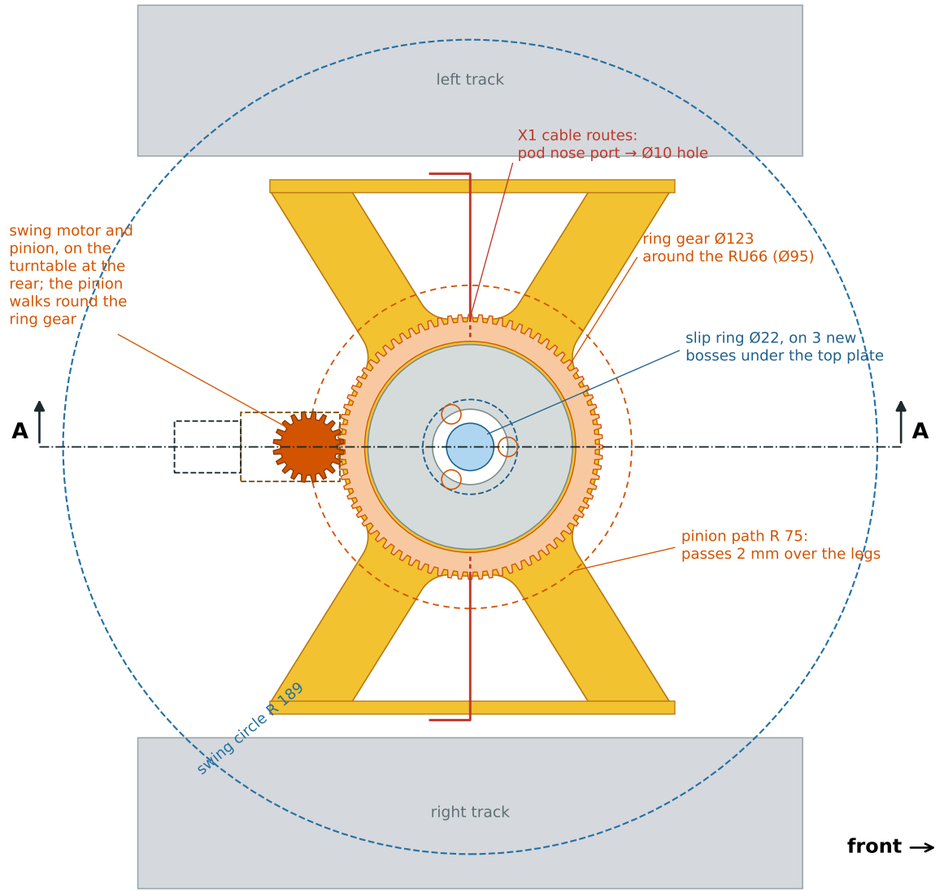
X1 turntable drive and wiring — concept for review

Concept sketch, not CAD. All dimensions in mm. Sheet 1 of 2: arrangement. X-frame, RU66 and tracks are X1 as built; orange = new printed parts, blue dashed = the future turntable. Both are proposals.

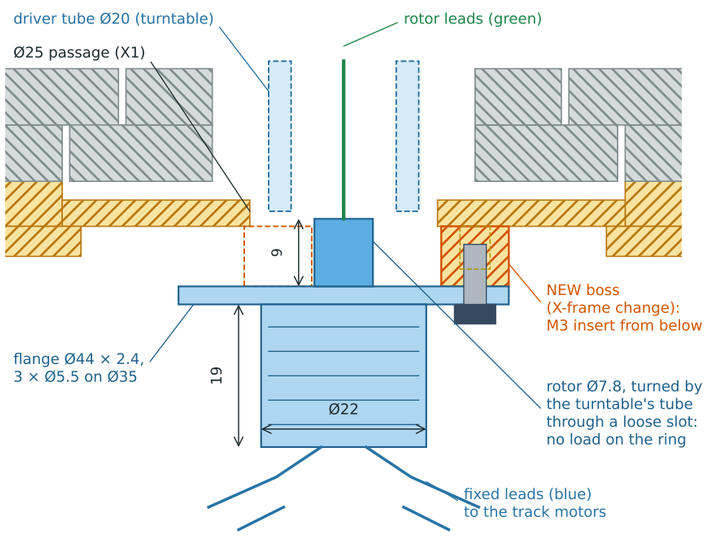
SECTION A–A on the centreline, seen from the right-hand side (front to the right)



PLAN VIEW, turntable removed (front to the right)



DETAIL C: slip ring at the swing centre (twice the scale of A–A)



How it turns

1. The ring gear is screwed to the X-frame and never moves, like the toothed race of a real swing bearing, which is fixed to the undercarriage.
2. The swing motor rides on the turntable. Its pinion walks round the ring gear and carries the whole upper with it. This is the full-size layout: on the Cat 320 the swing motor and its planetary reducer sit on the upper.
3. The RU66 carries the weight and the tipping loads of the upper. The gears only turn it.

Where the wires go

4. Battery, receiver and every motor driver stay in the upper body, as the engine, pump and valves do on the real machine. The battery also adds counterweight. The swing motor is up there too, so its wires never cross.
5. Only the two track motors are below the joint: 4 wires. They cross it through a slip ring at the swing centre, so the upper can turn any number of times without twisting a wire.
6. The slip ring's fixed half is screwed under the X-frame top plate. The turntable turns its rotor through a loose slot, so the slip ring carries no load. Its joints to the track-motor leads sit inside the carbody, behind the belly cover.

Changes to X1 that this needs

- 6 M3 inserts in the carbody top for the ring gear, and 3 bosses with M3 inserts under the top plate for the slip ring.
- The pinion's path (R 58 to 92, Z 87 to 100) joins the turntable reserve. Nothing on the chassis is in it now.

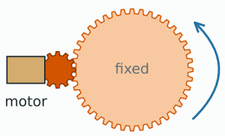
Sources: Cat 320 (07H) specifications p. 2; THK catalogue A18-20/21 (RU66); Prosper SRC022 series datasheet (SRC022A-6 = Adafruit 736); J9 model (JGY-370 envelope, Pololu #1999 hub). Sheet 2: options, sizing and decisions.

X1 turntable drive and wiring — options, sizing and decisions

Concept sketch, not CAD. Sheet 2 of 2. Proposals are marked; nothing here is modelled yet.

1 How the upper is turned

A. Pinion on a fixed ring gear, motor on the upper — proposed



The full-size layout (sheet 1). The motor can be either:

- JGY-370 worm gearmotor (proposed): self-locking, so the upper holds on a slope with no brake. Same part as the tracks. Shocks go into the teeth, so limit the current.
- Planetary gearmotor: the real machine's choice. Efficient and back-drivable, which is kind to the teeth, but it needs a brake or a holding current, or the upper drifts on a slope.

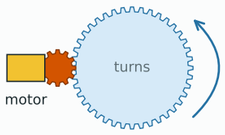
B. One big planetary set round the swing axis — not recommended



Sun gear on a motor at the centre, ring gear fixed to the X-frame, planet carrier = turntable. It works in principle, but:

- the sun gear and its motor sit exactly where the slip ring and the cables must pass;
- dozens of printed teeth must run true at Ø120 and more.

C. Motor on the chassis, ring gear on the upper — possible

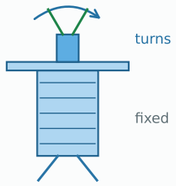


The 60 mm excavator's R3 layout: its slew motor sits on the lower bridge.

- Works, and keeps the turntable simple.
- But the swing motor's wires, and any limit-switch or angle-sensor wires, must also cross the joint: 6 or more circuits instead of 4.

2 Keeping the wires from tangling

1. Slip ring at the swing centre — proposed



Unlimited turning. Sliding contacts carry each wire across the joint. Needed: 4 circuits for the two track motors, plus 2 spare.

- SRC022A-6 (Adafruit 736): 6 rings, 2 A each, 240 V, up to 300 rpm, Ø22 × 19 body. The track motors stall at 1.3 A.
- Hobby RC excavators do this. The LESU PC30 turntable has a 4-way electric and 1-way oil slip ring and turns 360° without limit.

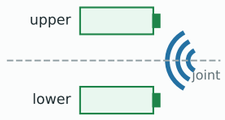
2. Limited swing with a cable loop — fallback



No slip ring: a loose loop of soft silicone wire round the centre, and end stops at about ±180°.

- Cheapest, and no sliding contacts.
- The upper cannot keep turning one way, so end stops or limit switches must stop it. The 60 mm excavator R3 used ±150° with no slip ring.

3. Own battery and receiver in the lower — possible



The lower half gets its own battery, receiver and track drivers.

- No wires cross the joint at all.
- Two batteries to charge and two receivers to set up, and the battery's weight no longer counts as counterweight.

3 Sizing (concept numbers)

	Cat 320	1:15, same density	Proposal A
Swing speed	11.25 rpm	—	7.5 rpm, no load
Swing torque	82 kN·m max	1.6 N·m	2.4 N·m rated

Torque scales with size⁴ at the same density: 82,000 / 15⁴ ≈ 1.6 N·m. Proposal A: JGY-370-1260 200:1 and gears 80 : 20 = 4 : 1.

JGY-370-1260, 12 V	Swing, no load	Swing torque, rated	
68 rpm, 87:1	17 rpm	1.1 N·m	too weak
40 rpm, 150:1	10 rpm	1.8 N·m	nearest the Cat 320
30 rpm, 200:1	**7.5 rpm**	**2.4 N·m**	**same as the tracks**
18 rpm, 340:1	4.5 rpm	4.7 N·m	slow, hard on teeth

Rated torques × 4, before gear losses. At stall the 200:1 motor gives 7.8 N·m at the swing; a 0.6 A current limit keeps it near 3.3 N·m, about 11 MPa in the printed teeth.

4 What full-size and RC excavators do

- Full size: a hydraulic swing motor with a planetary reduction gearbox is bolted to the upper frame. Its pinion meshes with the gear of the slewing ring, which is fixed to the undercarriage.
- A centre swivel joint on the swing axis carries the oil down to the travel motors, so the upper can turn freely.
- Hobby 1/14 models copy this in miniature. The LESU PC30 has a slewing support with a 1-way oil and a 4-way electric slip ring (360° without limit), and a brushless swing motor through a 603:1 reduction. The Huina 1580 has its own rotary gearbox between the slewing motor and the lower chassis.
- So a planetary gear set fits best as the reducer inside the swing gearmotor, as on the real machine (option A).

Key point on the wiring

Yes: the control circuit belongs in the upper. Keep the battery, receiver and every motor driver there. Then only the two track motors are below the joint, and just 4 wires have to cross it. A slip ring with 6 rings passes them, with 2 to spare, and the upper can turn without limit. Without a slip ring, the swing must be limited.

5 Decisions before I model it

Swing motor	JGY-370 worm, the track motor (proposed), or a planetary gearmotor?
Wires	Slip ring for unlimited turning (proposed), or limited swing with a cable loop?
Swing speed	7.5 rpm (200:1, proposed) or 10 rpm (150:1)? The Cat 320 swings at 11.25 rpm.
Then I model	the ring gear and pinion prints, the X-frame inserts and slip-ring bosses, the swing motor's place on the future turntable, and add the pinion's path to the reserve and to the checks.

Sources: Cat 320 (07H) specifications p. 2 (swing speed and torque); conequip.com "What is a Swing Drive?" (planetary gearbox, pinion, slewing ring on the undercarriage); oliverhobby.com LESU PC30 listing; havochobby.in Huina 1580 rotary gearbox listing; Prosper SRC022 series datasheet; excavator_60mm README (R3: ±150°, no slip ring).